classic type of diagram from the theoretical foundations of mathematics and computer science. In the context of blockchain, an FSM diagram is used to define the states and state transitions when a smart contract executes; it's a convenient tool for expressing the behavior of a smart contract.

> **DEFINITION** A *finite state machine* is made up of a set of states (an initial state and one or more terminal states), transitions from one state to another, and the events that bring about those transitions.

Let's explore the elements of the FSMm using an example problem and its finite state machine representation. We'll use the count coins use case from the vending machine use case diagram in section A.2.1 and design its logic in the form of an FSM. For simplicity, let's assume that the vending machine counts up to 25 cents and takes 5- and 10-cent coins as the only inputs. The FSM for counting to 25 cents is shown in figure A.3.

